



Outline

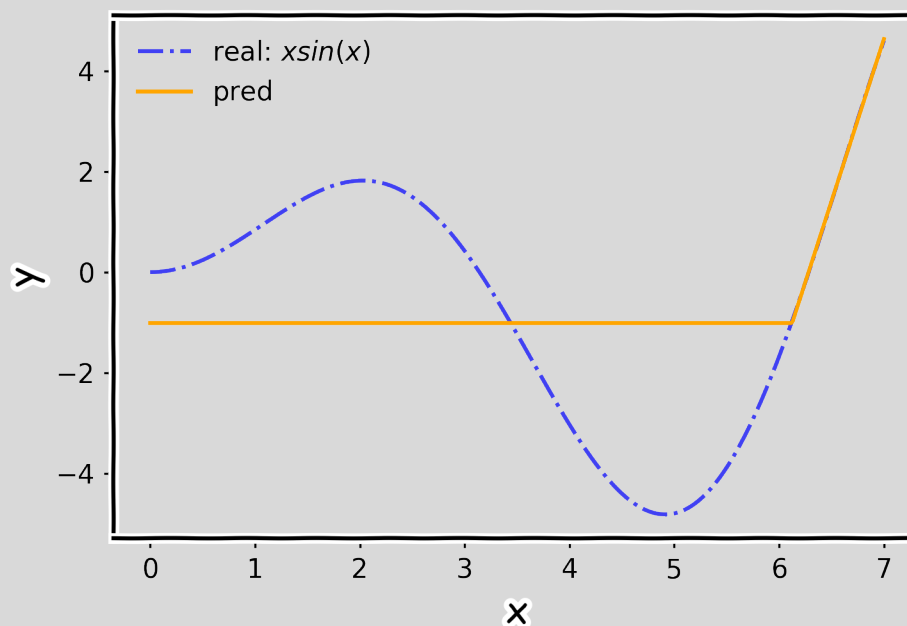
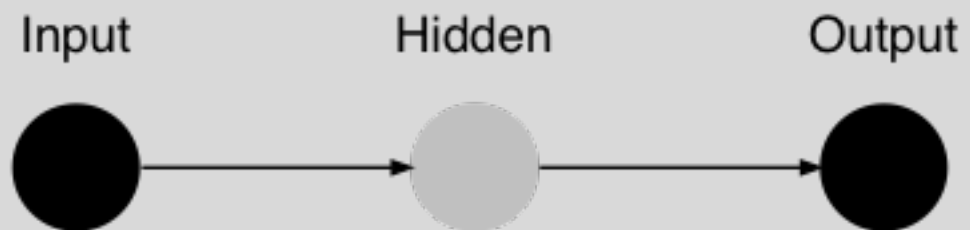
Anatomy of a NN

Design choices

- Activation function
- Loss function
- Output units
- **Architecture**

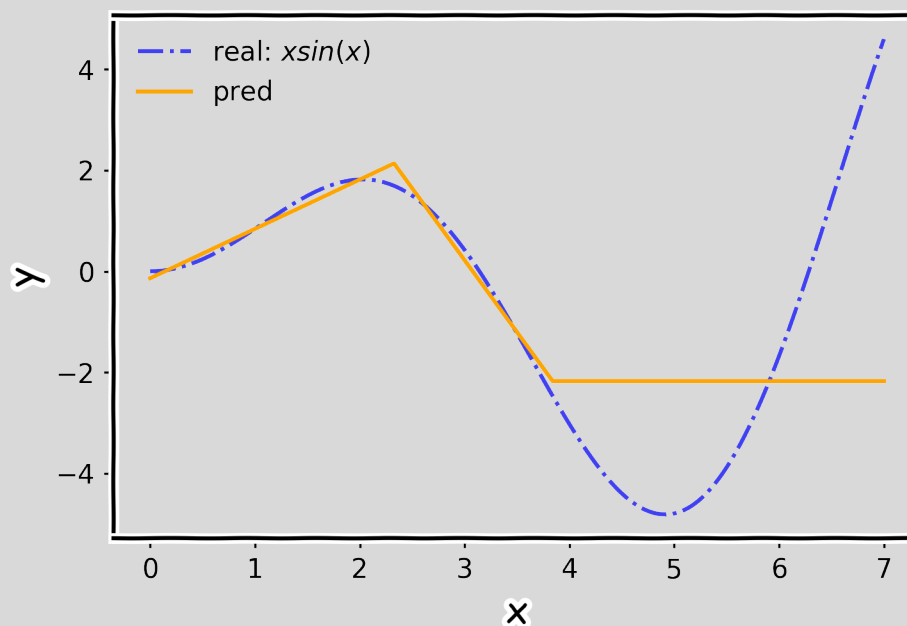
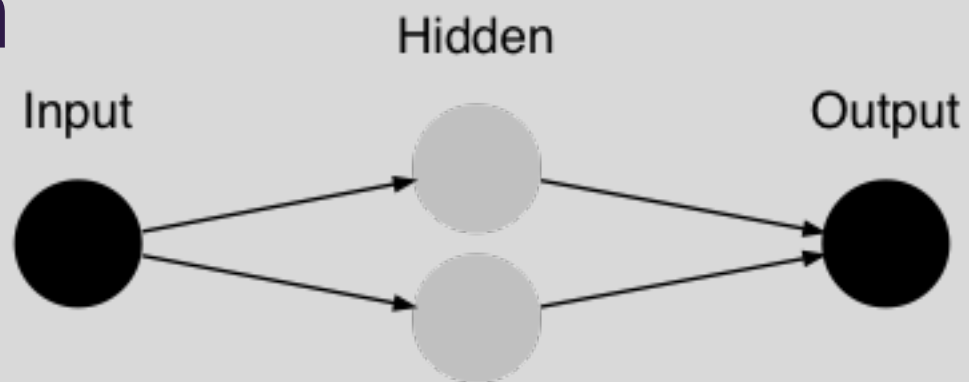


NN in action



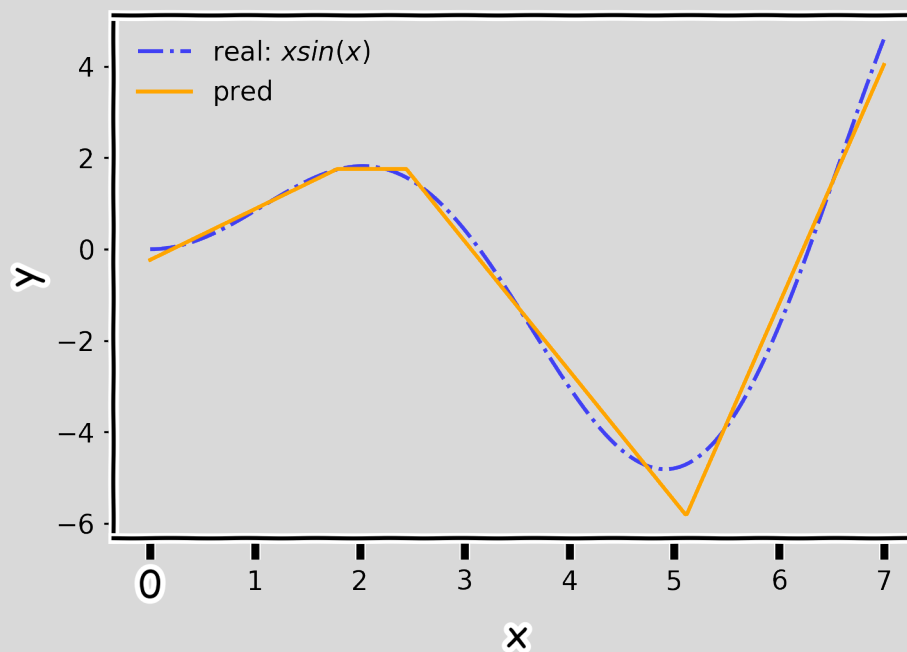
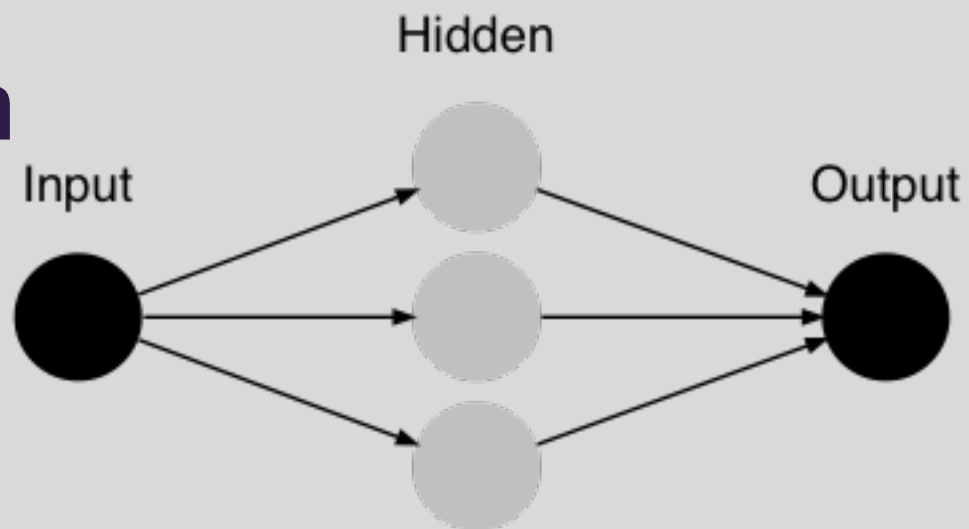


NN in action



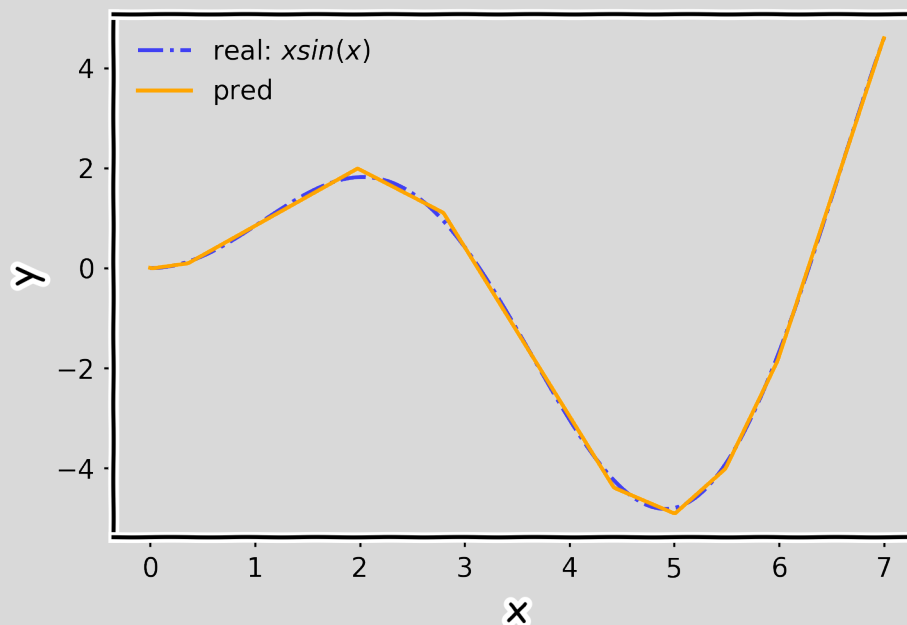
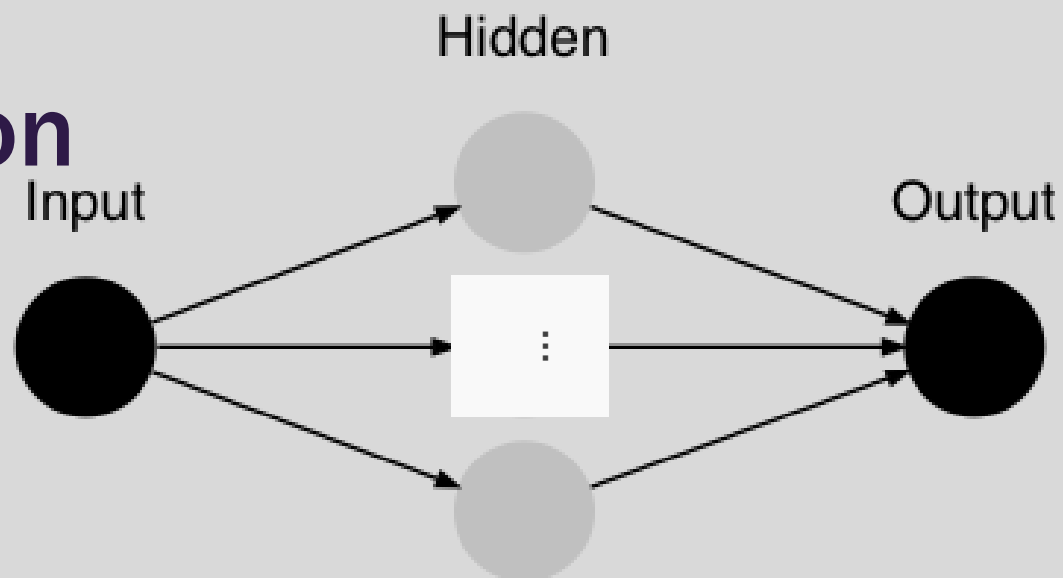


NN in action



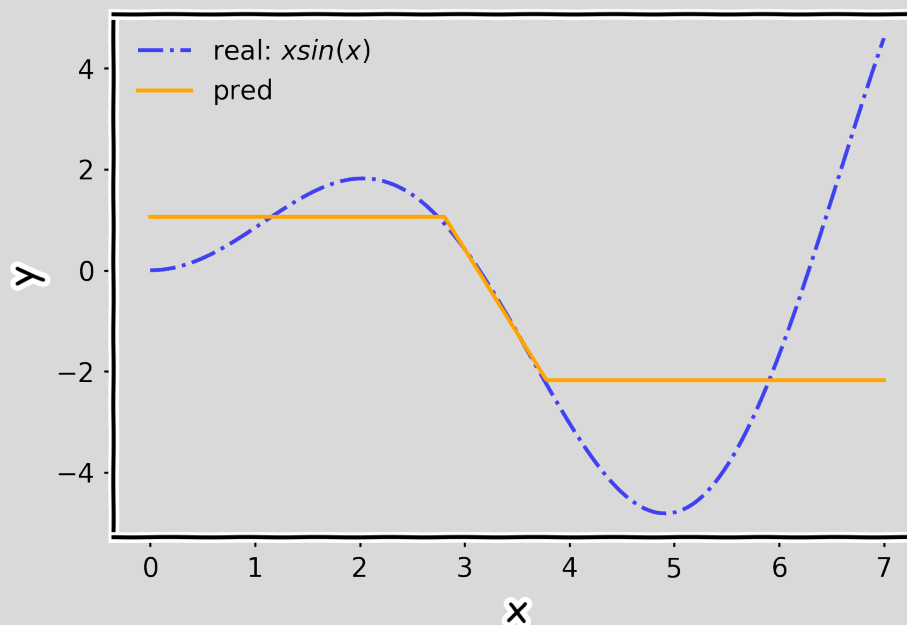


NN in action



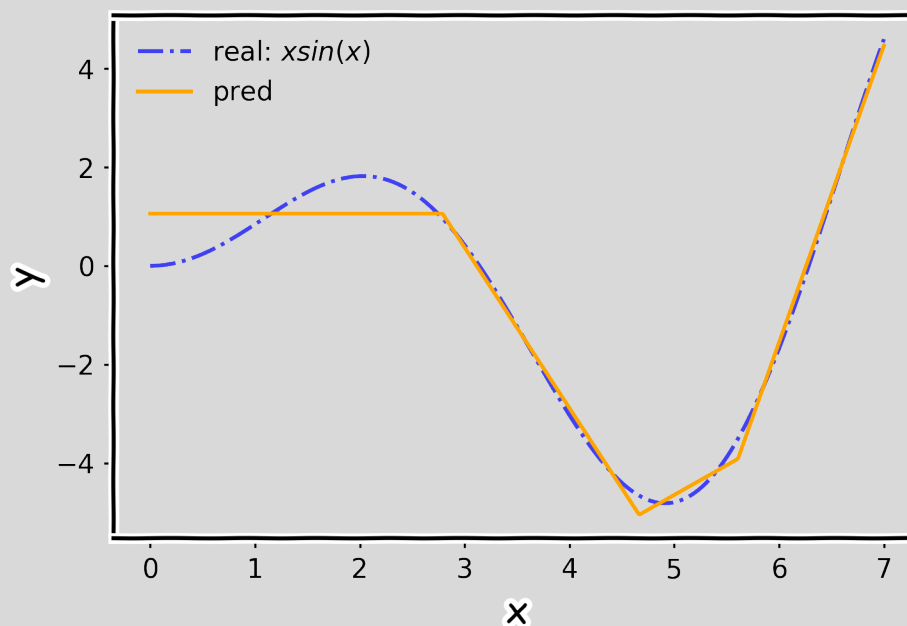
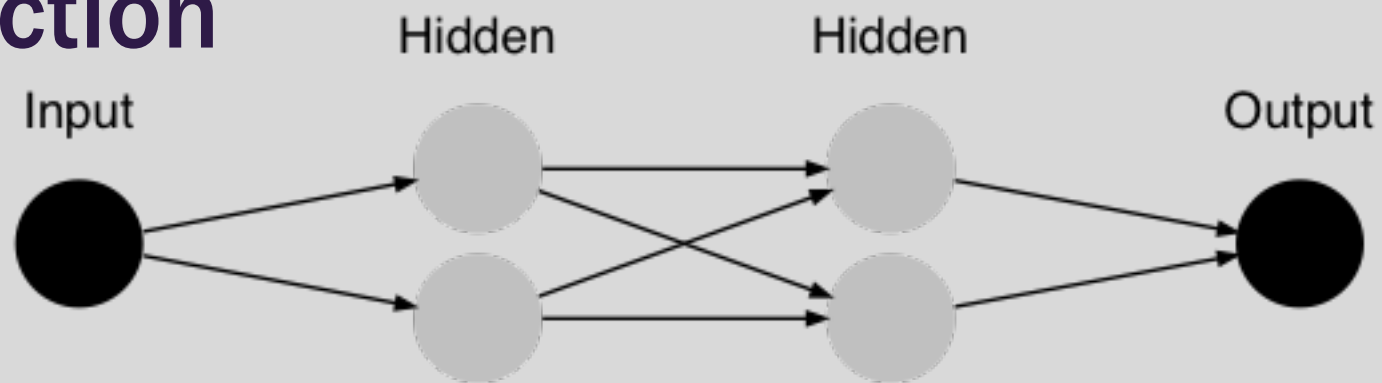


NN in action



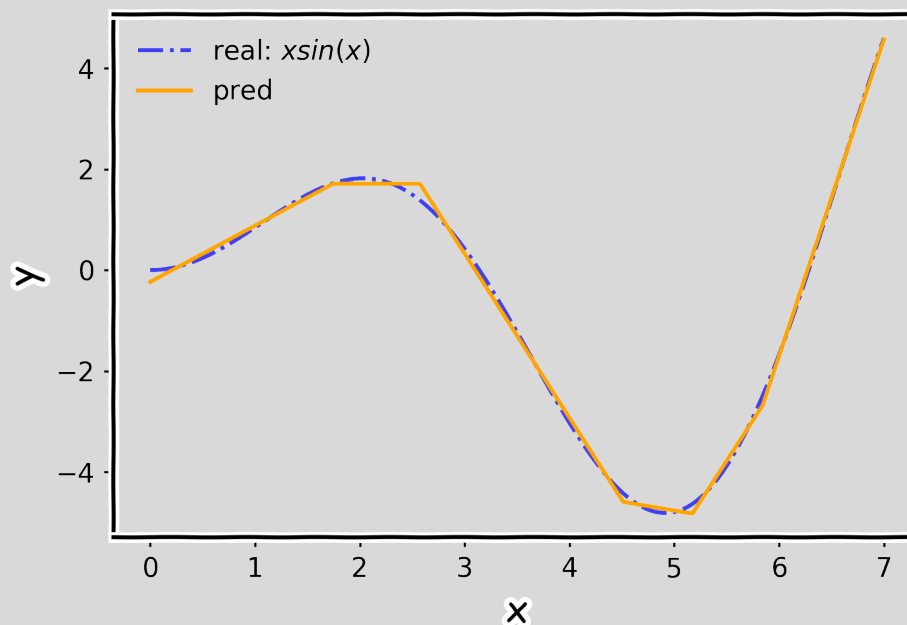
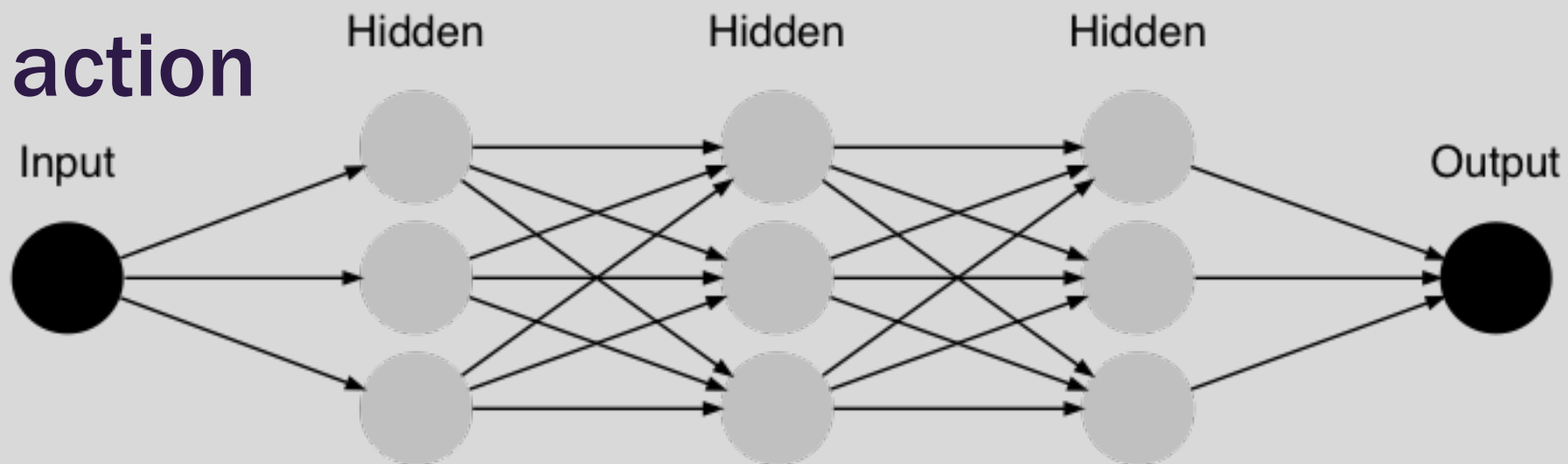


NN in action





NN in action





Universal Approximation Theorem

Think of a Neural Network as function approximation.

$$Y = f(x) + \epsilon$$

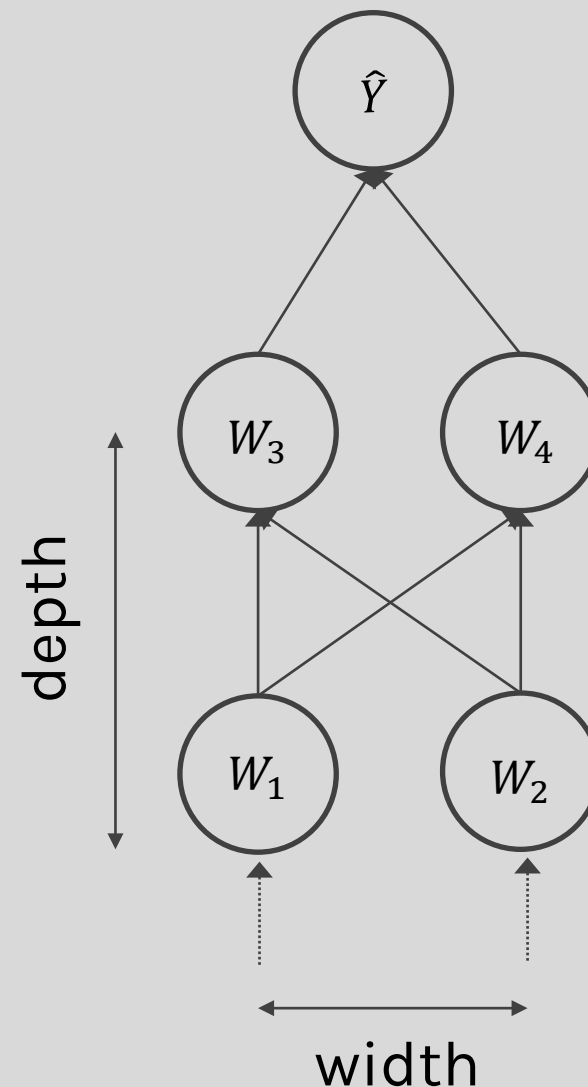
$$Y = \hat{f}(x) + \epsilon$$

$$\text{NN: } \Rightarrow \hat{f}(x)$$

One hidden layer is enough to *represent* an approximation of any function to an arbitrary degree of accuracy

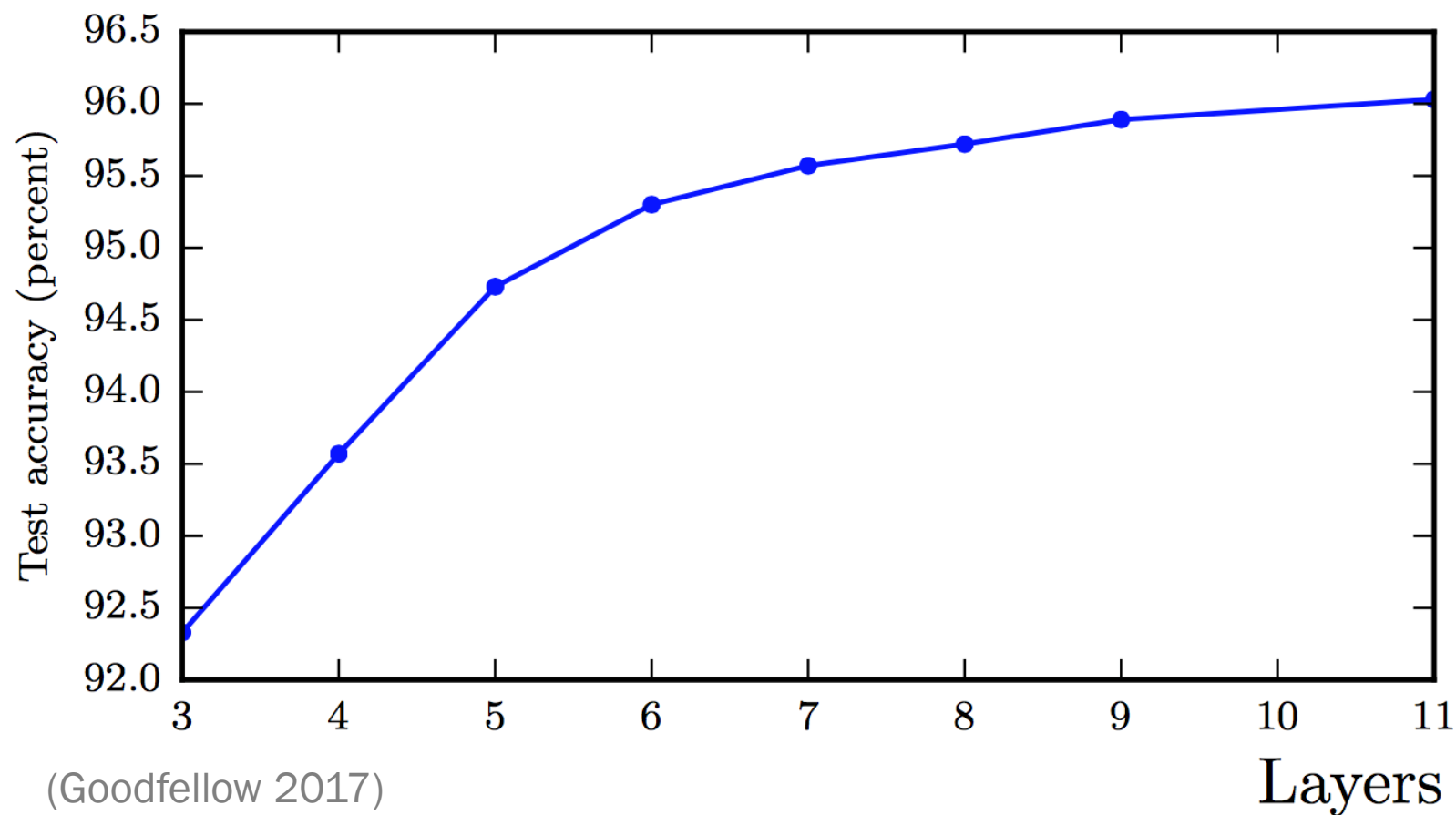
So why deeper?

- Shallow net may need (exponentially) more width
- Shallow net may overfit more





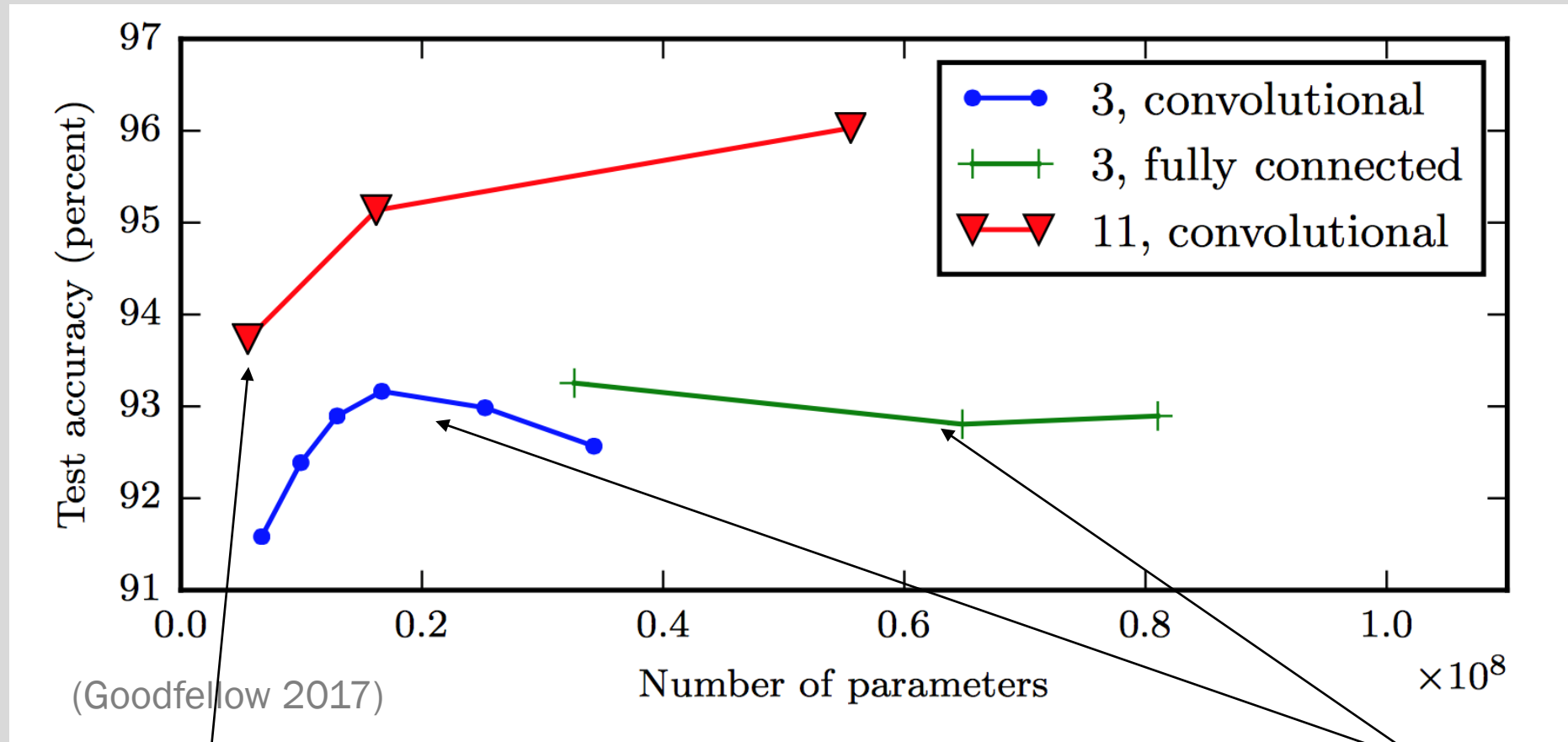
Better Generalization with Depth





Shallow Nets Overfit More

Depth helps, and it's not just because of more parameters



Don't worry about this word "convolutional". It's just a special type of neural network, often used for images.

The **11-layer net** generalizes better on the test set when controlling for number of parameters.

The 3-layer nets perform worse on the test set, even with similar number of total parameters.



Questions?



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